Damitri Kundu, Ph.D.

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Profiles LinkedIn Google Scholar GitHub ResearchGate

About me

Biostatistician with experience in early-phase oncology trials, specializing in dose optimization, Bayesian modeling, and analysis of complex longitudinal and survival data. Focused on developing statistically rigorous and computationally efficient solutions for real-world problems. (R, JAGS, clinical trials, predictive modeling)

Academic Experience

My Publications

- **Current Research**:
  - **PREDOSE : Pharmacometrically-Refined Early-phase Dose Optimization design for Oncology Study Enhancement**- Dr. Ayon Mukherjee (supervisor and project lead), Prof. Yuan Ji (*U. Chicago*), Dr. Damitri Kundu, Dr. Arunava Chakravartty and KyongJu Lee.
  - **Enhancing Restricted Mean Duration of Response for Interim Monitoring in Group Sequential Rare-Disease Cancer Trials**- Dr. Ayon Mukherjee (supervisor and project lead), Dr. Micheal Grayling, Dr. Damitri Kundu and Dr. James Wason.
- **Ph.D. thesis publications**:
  - **Kundu, D.**, Basu, S., Gogoi, M. and Das, K., 2025. A Bayesian latent class joint model for multivariate longitudinal and time-to-event data, *Journal of Applied Statistics*, pp. 1-17.
  - **Kundu, D.**, Krishnan, S., Gogoi, M. and Das, K., 2024. A Bayesian quantile joint modeling for multivariate longitudinal and event-time data, *Lifetime Data Analysis*, *30*(*3*), pp.680-699.
  - **Kundu, D.**, Sarkar, P., Gogoi, M.P. and Das, K., 2023. A Bayesian joint model for multivariate longitudinal and time-to-event data with application to ALL maintenance studies. *Journal of Biopharmaceutical Statistics*, *34*(*1*), pp.37-54.
- **Publications outside thesis**:
  - **Kundu, D.** and Das, K., 2024. A quantile-regression approach to bivariate longitudinal joint modeling, *Journal of Statistical Research*, *58*(*1*), pp.111-130.
  - Sen, S., **Kundu, D.** and Das, K., 2023. A flexible Bayesian variable selection approach for modeling interval data. *Statistical Methods & Applications*, *33*(*1*), pp.267-286.
  - Sen, S., **Kundu, D.** and Das, K., 2023. Variable selection for categorical response: A comparative study, *Computational Statistics*, *38*(*2*), pp.809-826.
  - Kedia, P., **Kundu, D.** and Das, K., 2023. A Bayesian variable selection approach to longitudinal quantile regression. *Statistical Methods & Applications*, *32*(*1*), pp.149-168.
- **Publications from Master's thesis**:
  - Das.S., **Kundu, D.** and Dewanji, A., 2022. Software reliability modeling based on NHPP for error occurrence in each fault with periodic debugging schedule, textitCommunications in Statistics-Theory and Methods, *51*(*14*), pp.4890-4902.

All my publications can be viewed at my ResearchGate and/or Google Scholar profiles (at the beginning of this document).

Ph.D. Experience

Aug, 2018–Dec, 23

Indian Statistical Institute, Kolkata

Key Areas: Bayesian modeling, joint modeling, survival analysis, dose optimization, R (JAGS)

- Successfully defended thesis titled **Bayesian approach to joint modeling of multivariate longitudinal and time-to-event data with applications to ALL maintenance studies** (Dec 2023).
- Developed novel Bayesian methods for joint modeling of longitudinal and survival data, with applications in clinical and biomedical studies.
- Implemented statistical models in R using JAGS for posterior inference and predictive analysis.

- Published **peer-reviewed papers** in statistical journals (see Publications section).
- Teaching Assistant for *Survey Sampling* (Coursera), managing 100+ students.
- Assisted in advanced training sessions at the *Advanced School for Statistical Learning*.

Industrial Experience

Associate Advisor - Statistics **Sep, 24–June, 26**

Eli Lilly and Company, Bangalore

Key Skills: R, R Shiny App, clinical trial methodology, Bayesian modeling, dose optimization and escalation, PK/PD-informed modeling, predictive analytics, industry–academia collaboration, project grant application, external conference presentations

- **Dose Optimization**:
  - ✓ *PREDOSE*, developing a Bayesian dose-optimization framework integrating patient-level PK/PD data within a seamless two-stage early-phase oncology trial design, aligned with **FDA Project Optimus**, under the supervision of Dr. Ayon Mukherjee.
  - ✓ Developed an R Shiny web application (PKComb-BOIN12 Simulator) supporting the manuscript *PKComb-BOIN12: A Pharmacokinetics-Informed Bayesian Optimal Interval Design for Dose Optimization in Cancer Drug-Combination Trials* (Dr. Jeiqi Tu, Dr. Ruitao Lin & Dr. Ayon Mukherjee, 2026 *manuscript in preparation*).
  - ✓ Developed an R Shiny web application (OBDC-Compare) and code based (Github: OBDC-Compare) supporting the manuscript *Optimal Dose Combination Selection in Oncology Trials Using Bayesian Designs: Statistical, Regulatory, and Operational Insights* (Dr. Ayon Mukherjee, Dr. Kentaro Takeda & Dr. James M.S. Wason, 2026 *manuscript in preparation*).
- **Lead Statistician**: Led statistical analyses in an *early-phase oncology* study, including authoring LoA, generating TFLs from raw data, contributing to Investigator Brochure submission, and providing statistical insights to clinical, medical, and data management teams through R-based analyses.
- **SMILe**: A dashboard (**S**tudy **M**anagement **I**nsights and **Le**arning) powered by PowerBI, with visualizations and statistical analyses generated in R, designed to predict various aspects of a clinical trial study, which contribute to decision making in the management of clinical trials:
  - ✓ **Enrolment Projection**: Bayesian modeling and bootstrap sampling in R JAGS were used to project enrollment and estimate additional patients needed to meet the target with >90% probability via *bisection* and *gradient descent* optimization.
  - ✓ **Evaluating Trends over Time**: Bayesian *posterior predictive distributions* were utilized to estimate and provide 95% CI for metrics AE, screening, IPD rate across various intervals of interest.
  - ✓ **Early Discontinuation Projection**: Applied *bootstrap sampling* together with survival analysis to estimate discontinuation probability and characterize patient dropout patterns across cohorts.

  Reduced computational runtime from **48 hours to 5 hours** across 218 studies in the SMILe dashboard by optimizing R code and leveraging high-performance libraries.
- **External Presentations**:
  - ✓ Presented published work on **Bayesian latent class joint modeling for multivariate longitudinal and time-to-event data** (*Journal of Applied Statistics*) at *Sankhya 2025* (Novartis Hyderabad) and *ConSPIC 2025* (Trivandrum).
  - ✓ Presented ongoing research on **PREDOSE: Pharmacometrically-Refined Early-Phase Dose Optimization** at *Conference on Data-driven Statistical Decision-making (CDSDM) 2026* (IIT Hyderabad, India).
  - ✓ International presentations: **PREDOSE: Pharmacometrically-Refined Early-Phase Dose Optimization**, accepted at *Statisticians in the Pharmaceutical Industry (PSI) 2026* (Belfast, UK) and *Adaptive Designs and Multiple Testing Procedures (ADMTP) 2026*, Newcastle University (Newcastle, UK), June 2026.
- **Industry–Academia Collaboration**:
  - ✓ Secured internal grant support for **PREDOSE**, enabling collaboration with Prof. Yuan Ji (*University of Chicago*).

✓ Conducted academic engagement sessions at Indian Statistical Institute (Kolkata) and Calcutta University.
✓ Contributed to proposed curriculum updates for ISI M.Stat program in *Survival Analysis* and *Clinical Trials*.

Senior Data Scientist **Aug, 23–Sep, 24**
Intuit, Bangalore

Key Skill: Machine learning modeling + GenAI + prompt engineering + data-driven decision support

Project Profitability: LightGBM-based modeling; **Product-Customer Engagement**: Time series analysis, Lasso, SHAP-based explainability; **Rare-Event Anomaly Detection**: Tree-based models with feature engineering; **Customer Query to SQL (GenAI)**: Multilabel classification using BERT and prompt engineering for NER.

✓ **WiDS Ambassador**: Conducted a workshop on *Automated Prompt Optimization in Large Language Models*, covering RAG-based methods (KATE, Skill-kNN) with practical demonstrations and cross-model comparisons.

Associate Data Scientist **Jun, 17– Aug, 2018**
RedBus, Bangalore

Workflow Prioritization: Sentiment analysis using multinomial logistic regression; **Hotel Classification**: Classification models using Random Forest; **Dynamic Pricing**: Regression-based modeling for pricing optimization; **Revenue Forecasting**: Time series models for expected marginal seat revenue.

Education

Ph.D. in Statistics **2018–2023**
Indian Statistical Institute, Kolkata
JRF, 2018 → SRF, 2020 → Thesis defended, 2023

Master's Degree in Statistics (M.Stat) **2015–2017**
Indian Statistical Institute, Kolkata
Statistics (Hons) : 73.2% [**First Division**]

Bachelor's Degree in Statstics (B.Sc) **2012–2015**
St.Xaviers College, Kolkata
Statistics (Hons) : 69.5%[**First Class**]

High School Degree **2010–2012**
Vivekananda Mission School
ICSE : 92% , ISC (Science and Maths) : 92%

Technical skills

Programming	R, RShiny, Python, PySpark, JAGS, BUGS, LaTeX
Statistical Methods	Bayesian inference, Bayesian variable selection, survival analysis, longitudinal data analysis, joint modeling, dose optimization, PK/PD modeling, MCMC methods
Machine Learning	Time series analysis, regression (Lasso, GLM), tree-based models (Random Forest, LightGBM), anomaly detection, model explainability (SHAP, LIME)
Tools & Platforms	R Shiny, PowerBI, Jupyter Notebook, MS Excel, LLMs (Claude, ChatGPT)
Libraries	dplyr, ggplot2, tidyverse, lme4, Rjags, mvJMBayes, survival, scikit-learn, TensorFlow, Keras, LightGBM
RShiny Apps	(i) PKComb-BOIN12, (ii) OBDC-Compare

Languages

English	First Language
Bengali	Native
Hindi	Third Language

Personal Interest

Painting